Understanding Derivatives for Nonlinear Functions

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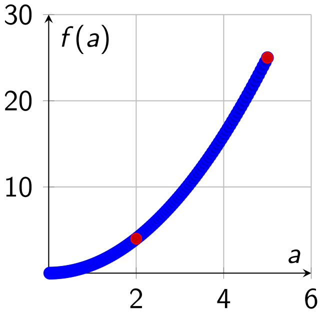
Recap

- For a linear function $f(a) = 3a$, the slope is constant (3) everywhere.
- Now, we explore functions whose slopes **change** at different points.
- Goal: build intuition for variable derivatives in nonlinear functions.

$f(a) = a^2$

- At $a = 2$: $f(a) = 4$, $f(2.001) \approx 4.004$ $\Rightarrow$ slope ≈ 4.
- At $a = 5$: $f(a) = 25$, $f(5.001) \approx 25.010 \Rightarrow$ slope ≈ 10.
- Derivative rule: $\dfrac{d}{da}a^2 = 2a$.

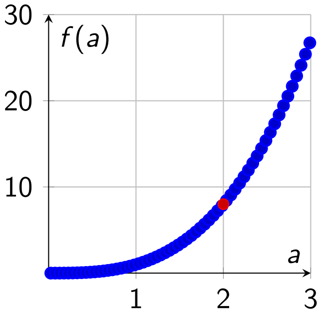


$f(a) = a^2$: Slope Varies

- The tangent slope increases as a increases.
- At $a = 2$: slope $= 4$; at $a = 5$: slope $= 10$.
- General rule: $\frac{d}{da}a^2 = 2a$ (from calculus tables).

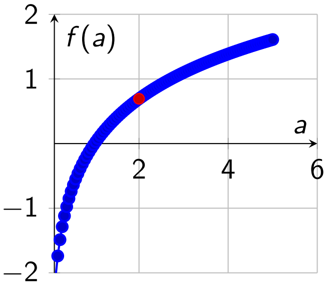
$f(a) = a^3$

- Derivative rule: $\dfrac{d}{da}a^3 = 3a^2$.
- At $a = 2$: slope $= 12$; small nudge $\Rightarrow$ $\Delta f \approx 12\,\Delta a$.
- Confirms: a^3 grows faster as a increases.

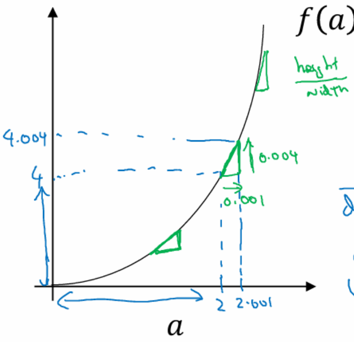


$f(a) = \log(a)$

- Natural log function: $f(a) = \ln(a)$.
- Derivative rule: $\dfrac{d}{da}\log(a) = \dfrac{1}{a}$.
- At $a = 2$: slope $= \frac{1}{2}$; for $\Delta a = 0.001$, $\Delta f \approx 0.0005$.
- $\Rightarrow$ The function grows slower as a gets larger.



Intuition about derivatives



$$\frac{d}{da} a^2 = 2a$$

0.001

$(2a) \times 0.001$

0.001

0.00000....01

$a = 2$ $\quad$ $f(a) = 4$

$a = 2.001$ $\quad$ $f(a) \approx 4.004$

$(4.004\boxed{001})$

slope (derivative) of $f(a)$ at $a=2$ is 4.

$\boxed{\frac{d}{da} f(a) = 4}$ when $\boxed{a=2}$.

$a = 5$ $\quad$ $f(a) = 25$

$a = 5.001$ $\quad$ $f(a) \approx 25.010$

$\boxed{\frac{d}{da} f(a) = 10}$ when $\boxed{a=5}$

$$\frac{d}{da} f(a) = \frac{d}{da} a^2 = \boxed{2a}$$

More derivative examples

$f(a) = a^2$ $\qquad \frac{d}{da} f(a) = \underbrace{2a}_{4}$ $\qquad a = 2 \quad f(a) = 4$ $\qquad a = 2.001 \quad f(a) \approx 4.004$

$f(a) = a^3$ $\qquad \frac{d}{da} f(a) = \underbrace{3a^2}_{3 \times 2^2 = 12}$ $\qquad a = 2 \quad f(a) = 8$ $\qquad a = \underline{2.001} \quad f(a) \approx 8.\underline{012}$

$f(a) = \log_e(a)$

$\ln(a)$

$\frac{d}{da} f(a) = \frac{1}{a}$

$\frac{d}{da} f(a) = \boxed{\frac{1}{2}}$

$a = 2 \quad f(a) \approx 0.69315$

$a = 2.\underline{001} \quad \underline{f(a)} \approx 0.69365$

$0.0005 \longleftrightarrow \underline{0.0005}$

Andrew Ng

Summary of Examples

Function	Derivative	Intuition
$3a$	3	Constant slope
a^2	$2a$	Slope grows with a
a^3	$3a^2$	Slope increases faster
$\log(a)$	$1/a$	Slope decreases as a grows

Takeaways

- Derivative = slope; for nonlinear functions, slope depends on where you evaluate it.
- Use calculus tables (or symbolic tools) for derivative rules.
- These principles form the foundation for **gradients in neural networks**.